

Fabio Pino

POSTDOCTORAL RESEARCHER ·

Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Wilberforce Rd, Cambridge CB3 0WA, United Kingdom

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Education

Université libre de Bruxelles (ULB) in partnership with VKI

Brussels, Belgium

DOCTOR OF PHILOSOPHY

Oct-19 - Mar-24

- Thesis: "Feedback control of liquid metal coating" Supervisors: Prof.Scheid (ULB) and Prof.Mendez (VKI)

The von Karman Institute for Fluid Dynamics (VKI)

Sint-Genesius-Rode, Belgium

RESEARCH MASTER IN FLUID DYNAMICS

Oct-18 - Jun-19

- Thesis: "Development of an adjoint method for receptivity analysis of ideal wing." Supervisor: Dr. Pinna
- Mark: **with honours**

Politecnico di Torino

Turin, Italy

2nd LEVEL SPECIALIZING MASTER IN SPACE EXPLORATION AND DEVELOPMENT SYSTEMS, SEEDS

Nov-17 - Sep18

- It aims at providing the students with the core applied physics and engineering skills required to develop space systems and missions from initial concept through design, assembly and verification, to launch and operations

Politecnico di Torino - Politecnico di Milano

Turin and Milan, Italy

ALTA SCUOLA POLITECNICA (ASP) DIPLOMA

Oct-14-Jun-16

- Restricted to 150 talented students, selected among the applicants to the Master of Science programmes of Torino and Milano Polytechnics. The ASP mission is to provide society with high-profile graduates combining in-depth disciplinary with interdisciplinary skills.

Politecnico di Milano

Milan, Italy

MASTER DEGREE IN AEROSPACE ENGINEERING

Sep-14 - Mar-17

- ASP students receive a Double Degree from Politecnico di Milano and Politecnico di Torino
- Mark: **110/110 with honours**

Politecnico di Torino

Turin, Italy

MASTER DEGREE IN AEROSPACE ENGINEERING

Sep-14 - Mar-17

- Thesis: "Study of the fluid flow and heat transfer in a power gearbox, using Fluent v17" - Supervisor: Prof.Pastrone
- Mark: **110/110 with honours**

Ecole centrale de Lyon

Lyon, France

DIPLÔME D'INGÉNIEUR GÉNÉRALISTE

Sep-12 - Sep-14

- He obtained a Double Degree (6 years - Top Industrial Managers for Europe (TIME) program

Politecnico di Torino

Turin, Italy

BACHELOR'S DEGREE IN AEROSPACE ENGINEERING

Sept-10 - Oct-14

- Thesis: "Development of software for the forecasting of the chill down of a cryogenic rocket engine" - Supervisor: Prof.Casalino
- Mark: **108/110**

Professional Experience

- Oct24 - **Postdoctoral Researcher**, Dep. of Applied Mathematics and Theoretical Physics (DAMTP), University of Cambridge - Cambridge, UK
- Present
- Mar24 - **Postdoctoral researcher**, Environmental and Applied Fluid Dynamics Department, The von Karman Institute for Fluid Dynamics - Sint-Genesius-Rode, Belgium
- Sep24
- Mar19 - **Postdoctoral researcher**, Transfers, Interfaces and Processes (TIPs) Laboratory, Universite libre de Bruxelles - Brussels, Belgium
- Sep24
- Oct19 - **Researcher PHD Student**, Environmental and Applied Fluid Dynamics Department, The von Karman Institute for Fluid Dynamics - Sint-Genesius-Rode, Belgium
- Mar24
- Oct19 - **Researcher PHD Student**, Transfers, Interfaces and Processes (TIPs) Laboratory, Universite libre de Bruxelles - Brussels, Belgium
- Mar24
- Nov22 - **Visiting PHD Student**, Dep. of Mathematics, Imperial College London - London, UK
- Apr23
- Jul18 - **Engineering Internship**, University of Leicester - Leicester, UK
- Sep18
- May18 - **Engineering Internship**, ISAE-SUPAERO - Toulouse, France
- Jul18
- Mar18 - **Graduate Internship**, Thales Alenia Space - Turin, Italy
- May18
- Sep16 - **Research Internship**, Gas Turbine and Transmissions Research Centre, University of Nottingham - Nottingham, UK
- Mar17
- Aug15- **Short Training**, Environmental and Applied Fluid Dynamics Department, The von Karman Institute for Fluid Dynamics - Sint-Genesius-Rode, Belgium
- Sep15
- May14- **Student Internship**, Snecma - Vernon, France
- Sep14
- July 2013 **Technical Internship**, Thales Alenia Space - Cannes, France

Publications

PUBLISHED

- Pino, F.**, Fracchia, E., Mendez, M. A., & Scheid, B. "Integral modelling and Reinforcement Learning control of 3D liquid metal coating on a moving substrate", *Physical Review Fluids*
- Pino, F.**, Mendez, M. A., & Scheid, B. (2024). "Multi-objective optimization of the magnetic wiping process in dip-coating", *Journal of Engineering Mathematics* 150.1 (2025): 20
- Pino, F.**, Mendez, M. A., & Scheid, B. (2024). "Linear stability analysis of a vertical liquid film over a moving substrate", *Journal of Fluid Mechanics* 1000 (2024): A57
- Pino, F.**, Scheid, B., & Mendez, M. A. (2024, June). Absolute/convective instability threshold in inverted falling film through linear stability analysis. In *AIP Conference Proceedings* (Vol. 3094, No. 1). AIP Publishing.
- Pino, F.**, Mendez, M. A., & Scheid, B. (2024). Absolute and convective instabilities in a liquid film over a substrate moving against gravity. *Physical Review Fluids*, 9(10), 104002.
- Pino, F.**, Schena, L., Rabault, J., & Mendez, M. A. (2023). "Comparative analysis of machine learning methods for active flow control." *Journal of Fluid Mechanics*, 958, A39.
- Ivanova, T., **Pino, F.**, Scheid, B., & Mendez, M. A. (2023). Evolution of waves in liquid films on moving substrates. *Physics of Fluids*, 35(1).
- Mendez, Miguel Alfonso, et al. "Challenges and opportunities for machine learning in fluid mechanics." *arXiv preprint arXiv:2202.12577* (2022).
- Vernicari, Pietro, et al. "Design and Optimization of Transfer of Resources from the Lunar Surface to Lunar Orbit." *Proceedings of the 69th International Astronautical Congress (IAC)*, Bremen, Germany. 2018.

Guardabasso, Paolo, et al. "Lunar outpost sustaining human space exploration by utilizing in-situ resources with a focus on propellant production." Proceedings of the 69th International Astronautical Congress (IAC), Bremen, Germany. 2018.

IN REVIEW

Pino, Fabio "Integral modelling of weakly evaporating 3D liquid film with variable substrate heating", Journal of Fluid Mechanics

IN PREP

Pino, Fabio, et al. "Linear feedback control of liquid film on upward-moving substrate. Part I. Modulation of free-surface stresses", Journal of Fluid Mechanics

Pino, Fabio, et al. "Linear feedback control of a liquid film on a moving substrate. Part II. Control using gas-jet actuation", Journal of Fluid Mechanics

Awards, Fellowships, & Grants

- 2025 **Winner of the 2025 ULB/VUB Syensqo Award**, Syensqo Fund
- 2025 **Winner – Affiliated Postdoctoral Membership Competition**, Clare Hall
- 2025 **Nova 111 List Italy Winner 2025**, Nova Talent Network
- 2025 **Finalist in the Belgian NCTAM Best PhD Thesis Award 2024**, Belgian National Committee for Theoretical and Applied Mechanics (NCTAM)
- 2024 **Finalist in the 19th ERCOFTAC Da Vinci Competition**, European Research Community On Flow, Turbulence And Combustion (ERCOFTAC)
- 2024 **Mobility and Congress funding**, Fonds de la Recherche Scientifique (FNRS)
- 2024 **Winner Postdoctoral fellowship**, Wiener Anspach Foundation
- 2023 **von Karman Institute (VKI)-Alumni Association Research Travel Grants 2023**, von Karman Institute Alumni association
- 2022 **Best presentation award session: Control Theory and systems**, 13th International Conference on Mechanical and Aerospace Engineering
- 2021 **Best Presentation Award PhDs Symposium**, von Karman Institute for Fluid Dynamics
- 2019 **PhD Scholarship - Bourses de Doctorat FRIA**, Fonds de la Recherche Scientifique - FNRS - Belgium
- 2018 **Scholarship from Compagnia di San Paolo for the the 2nd level Specializing Master in "Space Exploration and Development Systems"**, Compagnia di San Paolo
- 2017 **PEGASUS A.W.A.R.D., (Achievement through Working Abroad for academic Research or industrial Development projects)**, Politecnico di Torino
- 2014-2016 **Resident student at Collegio Einaudi**, Collegio Renato Einaudi Torino
- 2014 **Prix Francis Leboeuf**, Ecole Centrale de Lyon
- 2012 **Erasmus Scholarship for Double Degree Program**, Politecnico di Torino

Presentations

** presenting author; * mentored undergraduate*

CONTRIBUTED PRESENTATIONS

Pino, Fabio*. 2025. "Linear control strategies for liquid film on a substrate moving against gravity.". 2nd European Fluid Dynamics Conference (EFDC2), Dublin, Ireland.

Pino, Fabio*. 2024. "Absolute and convective instabilities in a liquid film over a substrate moving against gravity". 1st European Fluid Dynamics Conference (EFDC1), Aachen, Germany.

Pino, Fabio*. 2024. "LINEAR FEEDBACK CONTROL OF A LIQUID FILM OVER A SUBSTRATE MOVING AGAINSTS GRAVITY". 26th International Conference of the Theoretical and Applied Mechanics (ICTAM), Daegu, South Korea.

- Pino, Fabio***. 2023. "Linear stability analysis methods for flow system: Receptivity and Absolute/convective instability". Junior Fluid seminar Imperial College London, London, UK.
- Pino, Fabio***. 2023. "Absolute/convective instability threshold for a liquid film over a moving substrate via linear stability analysis". 16th European Coating Symposium, Paris, France.
- Pino, Fabio***. 2023. "Feedback machine learning control of 3D falling liquid film". 22nd Computational Fluid Conference, Cannes, France.
- Pino, Fabio***. 2022. "Absolute/convective instability threshold in inverted falling film through linear stability analysis". 20th International Conference of Numerical Analysis and Applied Mathematics, Eraklion, Greece.
- Pino, Fabio***. 2022. "Feedback machine learning control of 3D falling liquid film". 14th European Fluid Mechanics Conference, Athens, Greece.
- Pino, Fabio***. 2022. "Machine learning control of 2D and 3D falling liquid film". European Drag Reduction and Flow Control Meeting, Paris, France.
- Pino, Fabio***. 2022. "Comparison of white box and black box control methods of representative flow control test cases". 15th International Conference on Mechanical and Aerospace Engineering, Bratislava, Slovakia.
- Pino, Fabio***. 2022. "Comparative analysis of machine learning methods for active flow control". 8th European Congress on Computational Methods in Applied Sciences and Engineering, Oslo, Norway.
- Pino, Fabio***. 2021. "Modelling of magnetic wiping via integral model". 15th European Coating Symposium, Brussels, Belgium.

INVITED SPEAKER

- Pino, Fabio***. 2026. "Optimal substrate heating for enhanced heat transfer in phase-changing liquid films". Minisymposium: Interfacial Flows and Waves: Analysis, Modelling, and Computation; British Applied Mathematics Colloquium (BAMC) 2026, Norwich, United Kingdom.

Teaching Experience

- February 2023 **Hands-on Machine Learning for Fluid Dynamics 2023**, Lecutre Series von Karman Institute for Fluid Dynamics
- February 2022 **Hands-on Machine Learning for Fluid Dynamics 2022**, Lecutre Series von Karman Institute for Fluid Dynamics

Mentoring

- 2021-2022 **Omar Kahol**, Student of Alta Scuola Politecnica, Politecnico di Milano
- 2025 **Beatrice Losa**, Student of Alta Scuola Politecnica, Politecnico di Milano

Supervised Students

THE UNIVERSITY OF CAMBRIDGE

- 2024-2025 **Yusen Liang**, Third year summer internship, DAMTP, University of Cambridge

THE VON KARMAN INSTITUTE FOR FLUID DYNAMICS

2023-2024 **Edoardo Fracchia**, Master thesis internship, Politecnico di Torino
 2022-2023 **Aleksandr Ferrigno**, Master thesis internship, Politecnico di Torino
 2022-2023 **Francesco Margani**, Master thesis internship, Politecnico di Torino
 2021-2022 **Gabriele Gamba**, Master thesis internship, Politecnico di Torino
 2021-2022 **Riccardo Giacomino**, Master thesis internship, Politecnico di Torino
 2021-2022 **Balint Nyakas**, Research master project, von Karman Institute for Fluid Dynamics
 2020-2021 **Tsvetelina Ivanova**, Research master project, von Karman Institute for Fluid Dynamics
 2020-2021 **Andrea Palmarini**, Master thesis internship, Politecnico di Torino
 2020-2021 **Lorenzo Schena**, Master thesis internship, Politecnico di Torino
 2020-2021 **Mitja Desmet**, Internship, Vrije Universiteit Brussel (VUB)

Service and Outreach

REVIEWING

2026-
Present **Journal of Fluid Mechanics**, Reviewer

VOLUNTEERING

2025-2026 **Postdocs of Cambridge Society (PdOC)**, Organising Committee member
 2025-2026 **Clare Hall - Physics Special Interest Group**, Organizer
 2025-2026 **ERCOFTAC Course - Training course Surfing the ripple: fundamentals of liquid film dynamics**, Main Organizer
 2025-2026 **EUROMECH Colloquium 666 - Beyond the Free Surface: Liquid Film Flows from Theory to Applications**, Main Organizer
 2019-2022 **PhD Representative**, von Karman Institute for Fluid Dynamics
 2014-2015 **Student Representative**, Collegio Universitario di Torino "Renato Einaudi"
 2021-present **Student Mentor**, Alta Scuola Politecnica Alumni Association

Development

10th Complex Motion in Fluids summer school - University of Twente - Netherlands

10th Fluid Dynamics of Sustainability and the Environment (FDSE) Summer School - École Polytechnique - France

FLOW/INTERFACE School on Machine Learning and Data-Driven Methods 2019 - KTH Royal Institute of Technology - Sweden

Fluids under Control Summer School 2021 - Czech Technical University - Czech Republic